Lab 5 Static Code Analysis

R Suhaas | PES2UG24CS389 | Section G

Pylint output before fixes

Command: python -m pylint wildlife_sightings.py
Run date: 24 October 2026
Version: 4.1.3 | Python 3.12.4 | Exit code: 28

```
************* Module wildlife_sightings
wildlife_sightings.py:5:0: C0301: Line too long (103/100) (line-too-long)
wildlife_sightings.py:9:0: C0301: Line too long (117/100) (line-too-long)
wildlife_sightings.py:1:0: C0114: Missing module docstring (missing-module-docstring)
wildlife_sightings.py:8:0: C0116: Missing function or method docstring (missing-function-
docstring)
wildlife_sightings.py:8:0: C0103: Function name "logSighting" doesn't conform to snake_case naming
 style (invalid-name)
wildlife_sightings.py:8:0: W0102: Dangerous default value {} as argument (dangerous-default-value)
wildlife_sightings.py:8:16: W0621: Redefining name 'conn' from outer scope (line 4) (redefined-
outer-name)
wildlife_sightings.py:9:17: C0209: Formatting a regular string which could be an f-string
(consider-using-f-string)
wildlife_sightings.py:14:0: C0116: Missing function or method docstring (missing-function-
docstring)
wildlife_sightings.py:14:0: C0103: Function name "observerId" doesn't conform to snake_case naming
 style (invalid-name)
wildlife_sightings.py:18:0: C0116: Missing function or method docstring (missing-function-
docstring)
wildlife_sightings.py:18:0: C0103: Function name "zoneSummary" doesn't conform to snake_case
naming style (invalid-name)
wildlife_sightings.py:18:16: W0621: Redefining name 'conn' from outer scope (line 4) (redefined-
outer-name)
wildlife_sightings.py:20:4: R1705: Unnecessary "else" after "return", remove the "else" and de-
indent the code inside it (no-else-return)
wildlife_sightings.py:26:0: C0116: Missing function or method docstring (missing-function-
docstring)
wildlife_sightings.py:26:11: W0621: Redefining name 'conn' from outer scope (line 4) (redefined-
outer-name)
wildlife_sightings.py:32:0: C0116: Missing function or method docstring (missing-function-
docstring)

-----------------------------------
Your code has been rated at 3.46/10
```

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Pylint output after fixes

Command: python -m pylint --persistent=n wildlife_sightings.py
Run date: 24 October 2026
Version: 4.1.3 | Python 3.12.4 | Exit code: 0

```
--------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 3.46/10, +6.54)
```

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Bandit output before fixes

Command: python -m bandit -r wildlife_sightings.py
Run date: 24 October 2026
Version: 1.9.4 | Python 3.12.4 | Exit code: 1

```
Run started:2026-10-24 04:55:36.251883+00:00

Test results:
>> Issue: [B608:hardcoded_sql_expressions] Possible SQL injection vector through string-based
query construction.
   Severity: Medium   Confidence: Medium
   CWE: CWE-89 (https://cwe.mitre.org/data/definitions/89.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b608_hardcoded_sql_expressions.html
   Location: .\wildlife_sightings.py:9:17
8   def logSighting(conn, species, zone, count, extras={}):
9       conn.execute("INSERT INTO sightings (species, zone, count) VALUES ('{}', '{}',
{})".format(species, zone, count))
10      extras[species] = extras.get(species, 0) + count

--------------------------------------------------
>> Issue: [B324:hashlib] Use of weak MD5 hash for security. Consider usedforsecurity=False
   Severity: High   Confidence: High
   CWE: CWE-327 (https://cwe.mitre.org/data/definitions/327.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b324_hashlib.html
   Location: .\wildlife_sightings.py:15:11
14  def observerId(observer):
15      return hashlib.md5(observer.encode()).hexdigest()[:6]
16

--------------------------------------------------
>> Issue: [B101:assert_used] Use of assert detected. The enclosed code will be removed when
compiling to optimised byte code.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b101_assert_used.html
   Location: .\wildlife_sightings.py:36:4
35      logSighting(conn, "Hornbill", "East", 3, totals)
36      assert len(totals) == 3
37      print(observerId("Arun"), zoneSummary(conn, "North"), rarest(conn))

--------------------------------------------------

Code scanned:
    Total lines of code: 27
    Total lines skipped (#nosec): 0
    Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
    Total issues (by severity):
        Undefined: 0
        Low: 1
        Medium: 1
        High: 1
    Total issues (by confidence):
        Undefined: 0
        Low: 0
        Medium: 1
        High: 2
Files skipped (0):
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.12.4
```

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Bandit output after fixes

Command: python -m bandit -r wildlife_sightings.py
Run date: 24 October 2026
Version: 1.9.4 | Python 3.12.4 | Exit code: 0

```
Run started:2026-10-24 04:56:42.786656+00:00

Test results:
    No issues identified.

Code scanned:
    Total lines of code: 61
    Total lines skipped (#nosec): 0
    Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
    Total issues (by severity):
        Undefined: 0
        Low: 0
        Medium: 0
        High: 0
    Total issues (by confidence):
        Undefined: 0
        Low: 0
        Medium: 0
        High: 0
Files skipped (0):
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.12.4
```

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Flake8 output before fixes

Command: python -m flake8 wildlife_sightings.py
Run date: 24 October 2026
Version: 7.4.1 | Python 3.12.4 | Exit code: 1

```
wildlife_sightings.py:5:80: E501 line too long (103 > 79 characters)
wildlife_sightings.py:9:80: E501 line too long (117 > 79 characters)
wildlife_sightings.py:19:80: E501 line too long (93 > 79 characters)
wildlife_sightings.py:27:80: E501 line too long (96 > 79 characters)
wildlife_sightings.py:28:5: E741 ambiguous variable name 'l'
wildlife_sightings.py:28:6: E225 missing whitespace around operator
```

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Flake8 output after fixes

Command: python -m flake8 wildlife_sightings.py
Run date: 24 October 2026
Version: 7.4.1 | Python 3.12.4 | Exit code: 0

Flake8 produced no output. It completed with exit code 0, meaning no violations were reported.